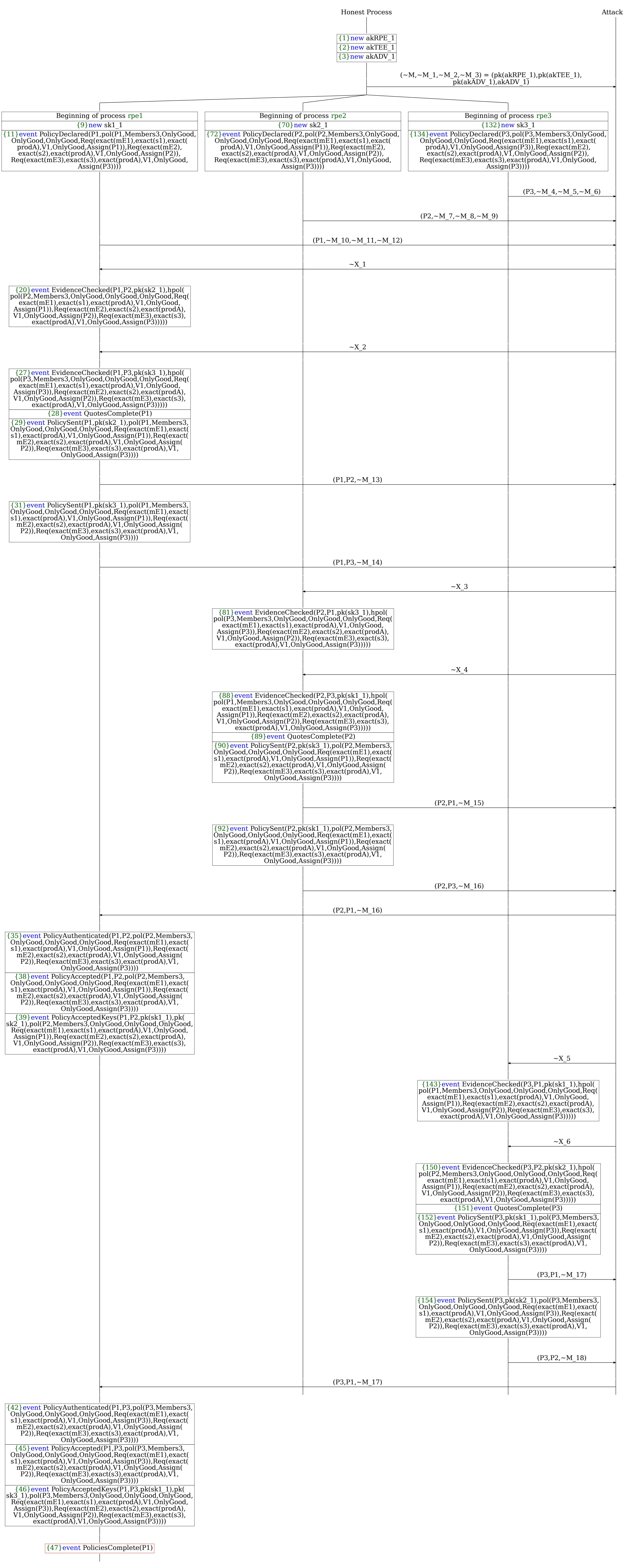
A trace has been found.

| Abbreviations |
|---|
| ~M_4 = sign(rpeQuote(mRPE,Good,hbind(pk(sk3_1),hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1) |
| ~M_5 = pk(sk3_1) |
| ~M_6 = hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))) |
| ~M_7 = sign(rpeQuote(mRPE,Good,hbind(pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1) |
| ~M_8 = pk(sk2_1) |
| ~M_9 = hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))) |
| ~M_10 = sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1) |
| ~M_11 = pk(sk1_1) |
| ~M_12 = hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))) |
| ~X_1 = (P2,~M_7,~M_8,hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))<br>= (P2,sign(rpeQuote(mRPE,Good,hbind(pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))) |
| ~X_2 = (P3,~M_4,~M_5,hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))<br>= (P3,sign(rpeQuote(mRPE,Good,hbind(pk(sk3_1),hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk3_1),hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))) |
| ~M_13 = sign(policyMsg(pk(sk2_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk1_1) |
| ~M_14 = sign(policyMsg(pk(sk3_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk1_1) |
| ~X_3 = (P1,~M_4,~M_5,hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))<br>= (P1,sign(rpeQuote(mRPE,Good,hbind(pk(sk3_1),hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk3_1),hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))) |
| ~X_4 = (P3,~M_10,~M_11,hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))<br>= (P3,sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))) |
| ~M_15 = sign(policyMsg(pk(sk3_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk2_1) |
| ~M_16 = sign(policyMsg(pk(sk1_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk2_1) |
| ~X_5 = (P1,~M_10,~M_11,hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))<br>= (P1,sign(rpeQuote(mRPE,Good,hbind(pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))) |
| ~X_6 = (P2,~M_7,~M_8,hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))<br>= (P2,sign(rpeQuote(mRPE,Good,hbind(pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))),akRPE_1),pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))) |
| ~M_17 = sign(policyMsg(pk(sk1_1),pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk3_1) |
| ~M_18 = sign(policyMsg(pk(sk2_1),pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))),sk3_1) |


Honest Process
Attacker
{1}new akRPE_1
{2}new akTEE_1
{3}new akADV_1
(~M,~M_1,~M_2,~M_3) = (pk(akRPE_1),pk(akTEE_1),pk(akADV_1),akADV_1)
Beginning of process rpe1
{9}new sk1_1
{11}event PolicyDeclared(P1,pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
Beginning of process rpe2
{70}new sk2_1
{72}event PolicyDeclared(P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
Beginning of process rpe3
{132}new sk3_1
{134}event PolicyDeclared(P3,pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P3,~M_4,~M_5,~M_6)
(P2,~M_7,~M_8,~M_9)
(P1,~M_10,~M_11,~M_12)
~X_1
{20}event EvidenceChecked(P1,P2,pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
~X_2
{27}event EvidenceChecked(P1,P3,pk(sk3_1),hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
{28}event QuotesComplete(P1)
{29}event PolicySent(P1,pk(sk2_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P1,P2,~M_13)
{31}event PolicySent(P1,pk(sk3_1),pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P1,P3,~M_14)
~X_3
{81}event EvidenceChecked(P2,P1,pk(sk3_1),hpol(pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
~X_4
{88}event EvidenceChecked(P2,P3,pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
{89}event QuotesComplete(P2)
{90}event PolicySent(P2,pk(sk3_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P2,P1,~M_15)
{92}event PolicySent(P2,pk(sk1_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P2,P3,~M_16)
(P2,P1,~M_16)
{35}event PolicyAuthenticated(P1,P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{38}event PolicyAccepted(P1,P2,pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{39}event PolicyAcceptedKeys(P1,P2,pk(sk1_1),pk(sk2_1),pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
~X_5
{143}event EvidenceChecked(P3,P1,pk(sk1_1),hpol(pol(P1,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
~X_6
{150}event EvidenceChecked(P3,P2,pk(sk2_1),hpol(pol(P2,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P1)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3)))))
{151}event QuotesComplete(P3)
{152}event PolicySent(P3,pk(sk1_1),pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P3,P1,~M_17)
{154}event PolicySent(P3,pk(sk2_1),pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
(P3,P2,~M_18)
(P3,P1,~M_17)
{42}event PolicyAuthenticated(P1,P3,pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{45}event PolicyAccepted(P1,P3,pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{46}event PolicyAcceptedKeys(P1,P3,pk(sk1_1),pk(sk3_1),pol(P3,Members3,OnlyGood,OnlyGood,OnlyGood,Req(exact(mE1),exact(s1),exact(prodA),V1,OnlyGood,Assign(P3)),Req(exact(mE2),exact(s2),exact(prodA),V1,OnlyGood,Assign(P2)),Req(exact(mE3),exact(s3),exact(prodA),V1,OnlyGood,Assign(P3))))
{47}event PoliciesComplete(P1)